Cuprous Oxide Cubic Particles with Strong and Tunable Mie Resonances for Use as Nanoantennas

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Supporting Information

I. Preparation of Cu_2O particles

Microemulsion method for Cu_2O spheres: The spherical Cu_2O particles were prepared using the water-in-oil microemulsion synthesis method. In this method, n-heptane, polyethylene glycol-dodecyl ether (Brij, average Mn ~362), copper nitrate, and hydrazine were used as the continuous oil phase, surfactant, copper precursor and reducing agent, respectively. These chemicals were added in the following sequence and quantity. First, 7.75 mL of n-heptane was added to three-neck round bottom flask at room temperature, followed by 1.15 mL of Brij surfactant. During the process of surfactant addition, the system was stirred vigorously, then 0.81 mL of 0.1 M copper nitrate aqueous solution was added, followed by 0.81 mL of 1 M of aqueous hydrazine solution. To understand the evolution of extinction spectra as a function of particle size, samples were taken at different time intervals for UV-Vis-near IR extinction measurements.

Chemical reduction method for Cu_2O cubes: The chemical reduction method reported in the literature was used for the synthesis of Cu_2O cubes.[1] Using this method, we first made 30 mL of 0.0032 M aqueous solution of $CuCl_2$, which acts as a copper source. This quantity of 30 mL is poured into a three-neck round bottom flask, which is placed under an inert atmosphere filled with nitrogen. To this solution at room temperature, we added 1 mL of 0.35 M aqueous NaOH solution, which is expected to result in $Cu(OH)_2$ colloids formation immediately.[1] Then, 1 mL of sodium ascorbate (reducing agent) was added. The solution then became brick-red, indicating the formation of Cu_2O particles.[1] The synthesis was done in aqueous solution at room temperature and nitrogen environment. The particles are dispersible in water, ethanol, and acetone. When left overnight in a vial, the particles settle at the bottom. However, the particles are easily re-dispersible by sonication.

II. Characterization of Cu_2O particles

Transmission Electron Microscope (TEM) Imaging: The TEM images were taken using JEOL JEM-2100 TEM and Thermo Fisher Scientific Titan Themis 200 G2 aberration corrected TEM. The JEOL JEM-2100 system is equipped with a LaB6 gun and an accelerating voltage of 200 kV. The Titan Themis 200 system is equipped with a Schottky field-emission electron gun and operated at 200 kV. Samples were prepared by taking 150 µL of the sample into 2 mL of deionized water. This dilution solution was sonicated to ensure proper dispersion. Next, 10 µL was taken and placed onto a TEM grid with ultra-thin carbon supporting film. The water was allowed to evaporate at room temperature before TEM imaging.

UV-Vis-near IR extinction measurements: All UV-Vis-near IR extinction spectra were taken using an Agilent Cary 60 Spectrophotometer. 150 µL of the synthesis mixture was diluted into 2 mL of deionized water. This diluted sample was then used for UV-Vis-near IR extinction measurements.

X-ray diffraction (XRD) patterns: XRD patterns were acquired using a Philips X-Ray diffractometer (Phillips PW 3710 mpd, PW2233/20 X-Ray tube, Copper tube detector – wavelength - 1.5418 Angstroms), operating at 45 KW, 40 mA. Figures S1a-b show the representative XRD spectra, which conform to the Cu_2O features, as reported in our previous contributions.[2,3]

III. Details of finite-difference time-domain (FDTD) Simulations

To implement FDTD simulations, we employed the Lumerical FDTD package.[4] The optical properties of Ag, and Cu_2O were taken from Palik.[5] The real (n) and imaginary (k) parts of the refractive index used in the simulations for Cu_2O are shown in Table S1. The boundary conditions used for the simulations were perfectly matched layer (PML) conditions in the x, y, and z directions. For the simulations of extinction, scattering, and absorption spectra, the respective cross sections as a function of wavelengths were calculated using the total-field/scattered-field (TFSF) formalism. The incident light source used for these simulations was the Gaussian source in the simulated wavelength region. For the simulations of the magnetic and electric field distributions, a plane wave was used for electromagnetic field incidence with propagation in the x-axis direction, and polarization along the y-axis and the z-axis for the electric field and the magnetic field, respectively. For all the simulation results shown in the main draft, and the directly associated supporting figures (i.e., Figures S3a-d, S4a-d, S5a-d, S6a-d, and S7-S13), the simulations were performed for Cu_2O articles in a surrounding medium with the refractive index of 1. We also simulated the effect of the refractive index of the medium on the Mie resonance of Cu_2O particles. For these simulations, we used the refractive index of water (~1.33). Figure S14 and Figure S15 show the representative simulation results with the medium refractive index of 1.33. By the comparison of Figure S4a-d (i.e., refractive index of 1) and Figure S15a-d (i.e., refractive index of 1.33), it can be seen that the magnetic field and electric field enhancement values are damped for the refractive index of 1.33.

IV. Single particle scattering acquisitions

The scattering spectrum of a single particle was acquired using a Renishaw RM 1000 micro-Raman spectrometer, which is customized with a 150 lines/mm grating for broadband measurements. The radiation was collected using a 100× objective lens (NA = 0.85) integrated with a dark-field condenser (Renishaw). A 275 W halogen bulb (Osram HLX 64656, 24 V) was employed as the light source (operated at 20 V AC). Confocal collection of the signal was achieved by a virtual pinhole by the crossover of the slit (set to 60 μm) and a 7 pixel rows (7×22 = 154 μm wide) on the CCD. The size of a pixel is 22×22 μm^2.

Because the wavelength range of a single scan was not sufficiently wide for the Cu_2O cubes, each spectrum was acquired from two scans, performed in adjacent spectral regions. The adjacent spectra were stitched together at 738 nm. The spherical Cu_2O particles were scanned in a single wavelength range. A holographic 532 nm edge filter (Semrock) was employed to reject higher order diffractions from the grating.

The "scattering intensity (a.u.)" spectra of Figures 7 and 8, $S(\lambda)$, is computed as:

$$S(\lambda) = \frac{(S^{*}(\lambda) - T(\lambda)) - k(B(\lambda) - T(\lambda))}{R(\lambda) - T(\lambda)}$$

where, $S^*(\lambda)$, $R(\lambda)$, $T(\lambda)$, $B(\lambda)$, k, are the raw (uncorrected) signal from the NP, reference signal from the white calibration standard (Spectralon by Labshpere), thermal (dark) signal from the thermoelectric-cooled CCD detector, background signal collected from the substrate surface, and a correction factor (varied from 0.96 to 1.00) for the background signal, respectively. The particle and the background were integrated for 200 s and the white calibration standard was integrated for 108 s, all in dark field optics. All acquisitions were performed twice per particle to confirm repeatability.

Submicron particles, suspended in ethanol, were spotted in 5 μL aliquots onto a 23 mm × 23 mm cleaned microscope grid-slides (Bellco Glass, Inc.) using a micropipette. This suspension was obtained by 5-min ultrasonication of the as-synthesized colloid after it was 100×-diluted in ethanol. After drying of the solvent, a single particle was selected from its diffraction-limited dark field image. Optimum positioning of the particle in the focal point of the lens took about 5-10 min until drifting decelerated to a negligible rate. Prior to the acquisition, the signal image on the CCD detector was checked to confirm signal is fully detected and binning is minimized for highest signal-to-noise.

Succeeding the scattering acquisitions, the measured particles were imaged by scanning electron microscopy (SEM) for shape and size determination and single-particle confirmation. To trace and locate the particles correctly, the position of the particles on the microscope grid-slides were recorded with microscope camera using 100×, 50× and 5× lenses. Prior to SEM, the samples were coated with carbon to avoid charging. SEM was performed with a FEI Quanta 600 field-emission microscope, operated at 30 kV.

V. Calculation of the scattering cross section

Scattering cross section of a submicron Cu_2O cube was calculated by normalization of the scattering counts with diffused reflectance counts from a 3.5 μm diameter polytetrafluoroethylene (PTFE) bead (at the same integration time of 200 s) and subsequently multiplying the normalized value with the surface area of the spherical PTFE microbead. PTFE was employed due to its stability under bright microscope illumination and near-unity diffused reflectivity. To capture all the light, diffuse-reflected from the PTFE bead, we used a slit entrance of 500 μm and 30-pixel rows on the CCD detector. Under dark field illumination, the diffused-reflection occurs from the upper hemispherical area of the bead. However, the objective lens also collects only about half of the scattered radiation from a submicron Cu_2O particle. Only the light scattered to the far field in the semi-infinite space around the particle (above the substrate) is collected. Therefore, scattering counts must be corrected with a multiplication factor of 2 to take into account the total scattering when reporting the correct value of the cross section for a particle. Equivalently, in the above calculation, we multiply the normalized counts of the submicron particle with the full surface area of the PTFE standard sphere (instead of hemispherical surface area).

VI. Single-particle Raman measurements

The crystal structure of the nanocubes were studied by single-particle Raman spectroscopy with a WITec alpha 300R Raman microscope, equipped with a 532 nm laser. Raman spectrum of a single nanocube was collected using a 100× objective lens of 0.9 numerical aperture, 600 lines/mm grating, and 100 μm confocal aperture (fiber) diameter. The incident laser power and spot size were set to 0.052 mW and diffraction limited (minimum), respectively. The signal was integrated for 100 s. The laser power was optimized (maximized) to the onset of spectral shifting, caused by laser heating.

Table S1. The refractive index values of Cu_2O used in the simulations.[5]

	Cu_2O	
Wavelengths (nm)	**n**	**k**
300	2	1.85
350	2.4	1.44
400	2.8	0.99
450	3.06	0.6
500	3.12	0.35
550	3.1	0.19
600	3.02	0.13
650	2.9	0.1
700	2.83	0.083
750	2.77	0.07
800	2.7	0.06
850	2.66	0.053
900	2.63	0.048
950	2.61	0.043
1000	2.6	0.04
1100	2.59	0.033
1200	2.58	0.027
1300	2.57	0.021
1400	2.57	0.017
1500	2.57	0.013
2000	2.56	0.002

X-ray diffraction patterns:

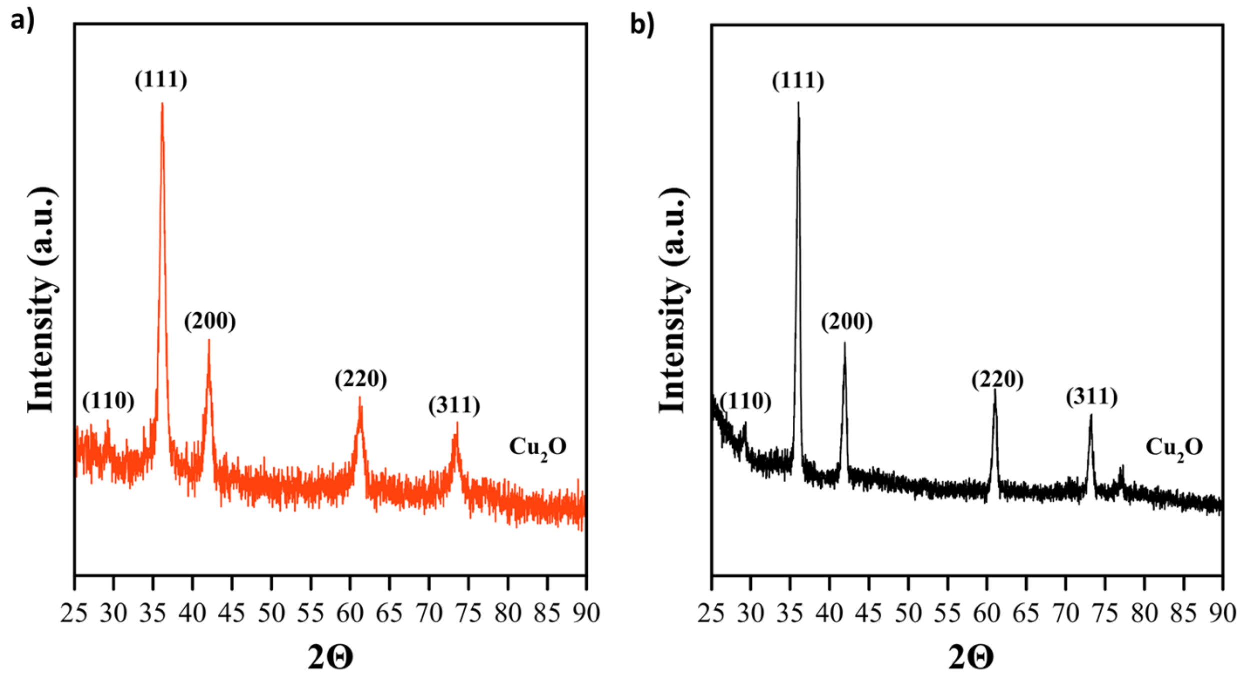

Figure S1. The representative X-ray diffraction pattern of Cu_2O **(a)** quasi-spherical and (**b)** cubical particles.

Table S2. TEM-measured average particle sizes of Cu_2O spheres during growth synthesis at different time intervals.

Time (hr)	**Average size (nm)**
1	5.3 ± 1.7
4	120 ± 22
8	125 ± 28

Table S3. TEM-measured average particle sizes of Cu_2O cubes during synthesis at different times.

Time (min)	**Average size (nm)**
2	19 ± 2
5	267 ± 83
75	327.3 ± 125

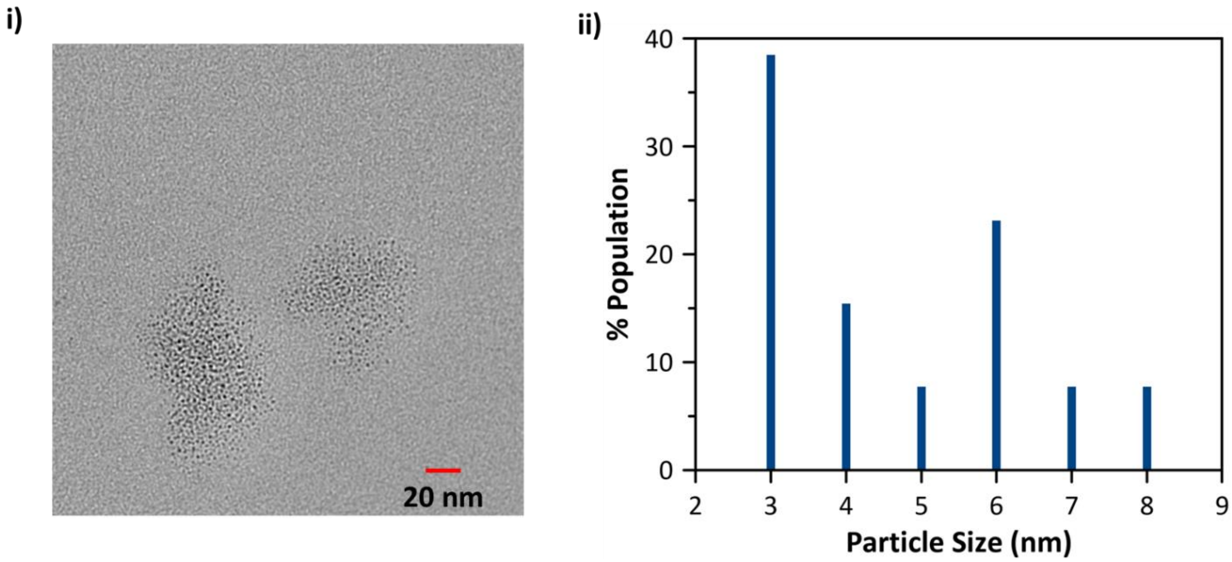

Figure S2a. i) Representative TEM image of particles, and **ii)** size distribution of particles, for the sample collected at 1 hr during the synthesis of Cu_2O spheres. Average size of particles = 5.3 ± 1.7 nm.

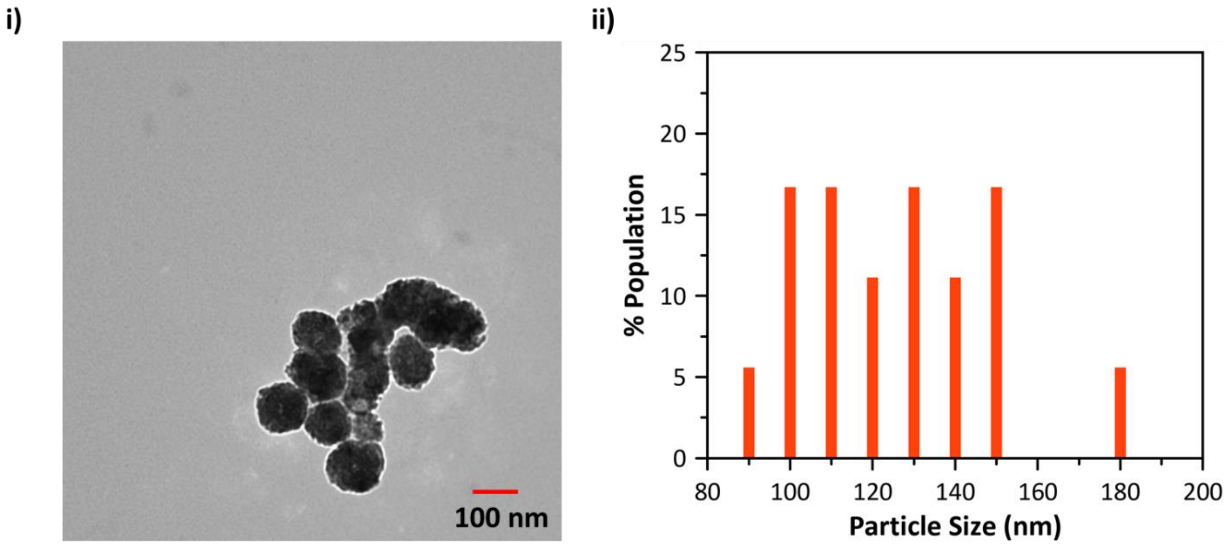

Figure S2b. i) Representative TEM image of particles, and **ii)** size distribution of particles, for the sample collected at 4 hr during the synthesis of Cu_2O spheres. Average size of particles = 120 ± 22 nm.

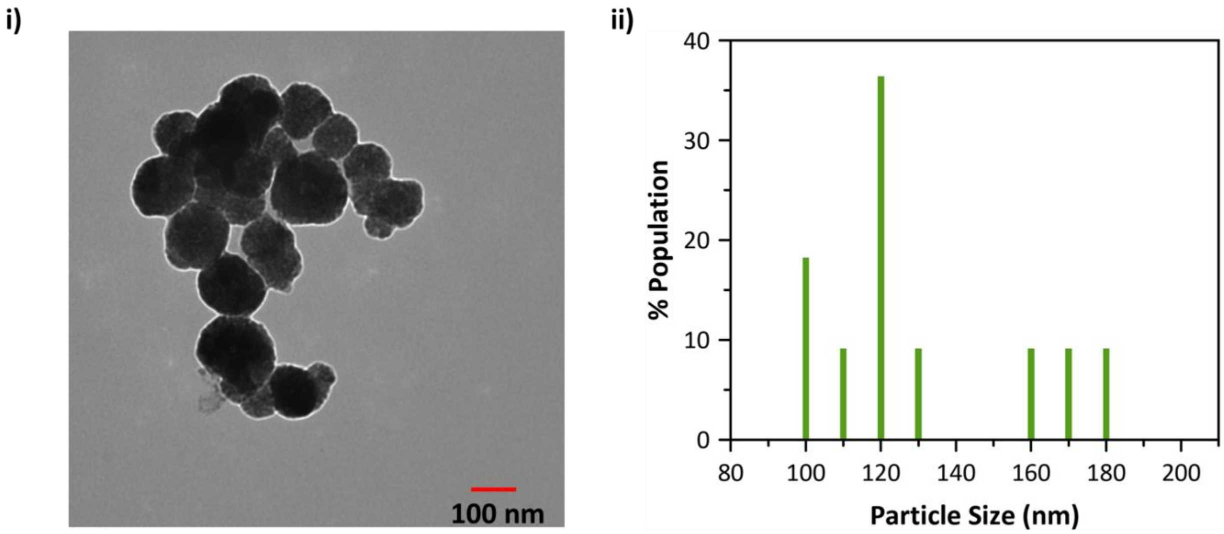

Figure S2c. i) Representative TEM image of particles, and **ii)** size distribution of particles, for the sample collected at 8 hr during the synthesis of Cu_2O spheres. Average size of particles = 125 ± 28 nm.

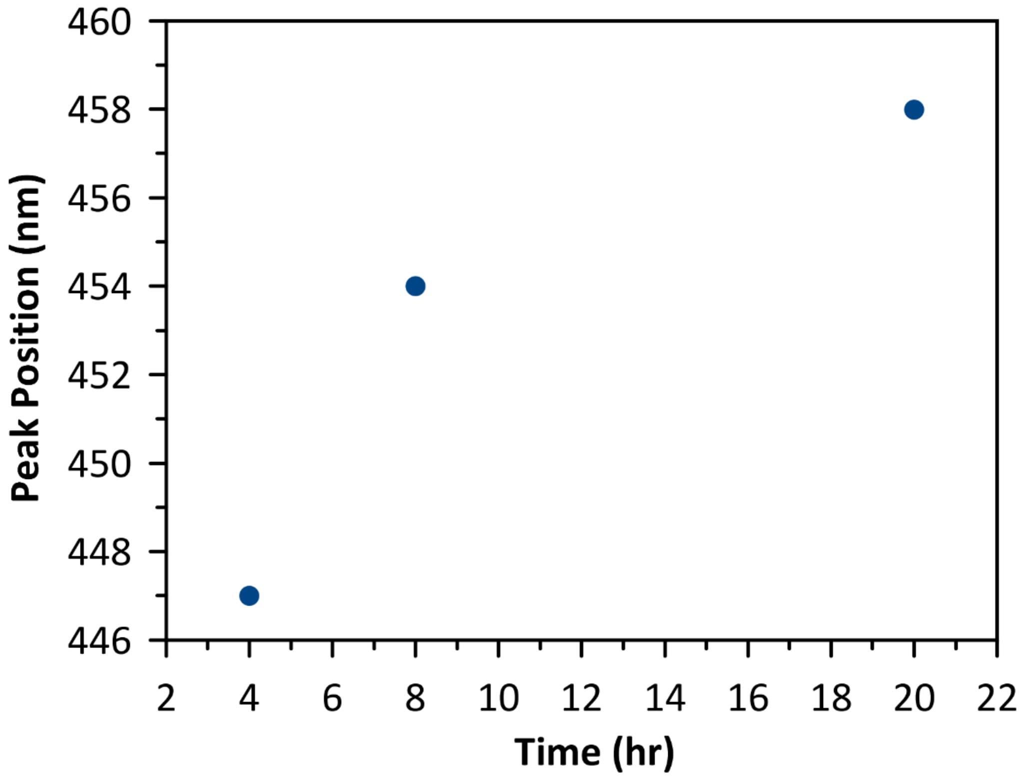

Figure S2d. Experimentally measured extinction peak position (nm) of Cu_2O spherical particles as a function of synthesis time.

i) ii)

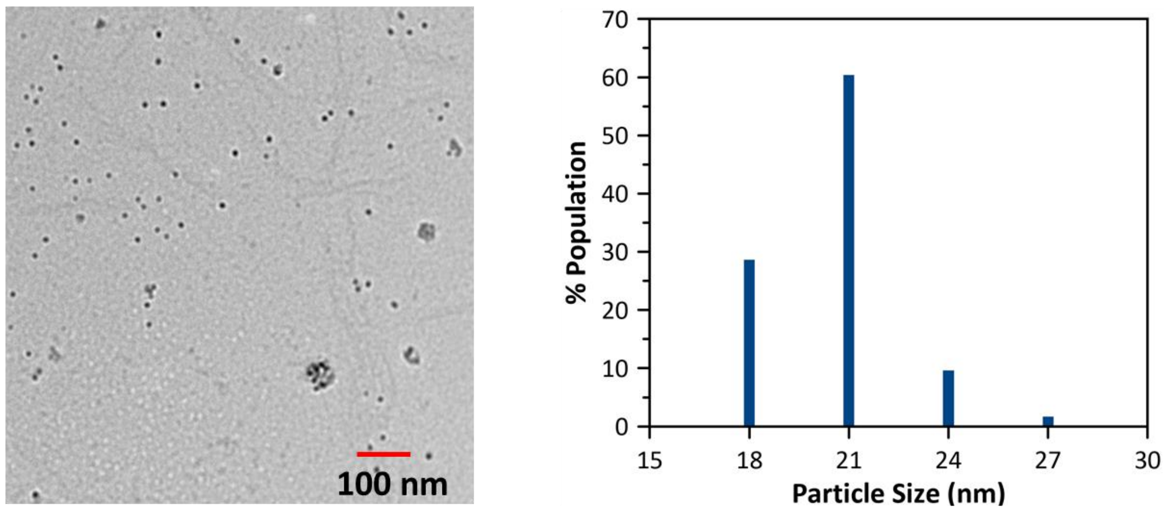

Figure S2e. i) Representative TEM image of seed particles, and **ii)** size distribution of the corresponding particles, for the sample collected at 2 minutes during the synthesis of Cu_2O cubes. Average size of particles = 19 ± 2 nm.

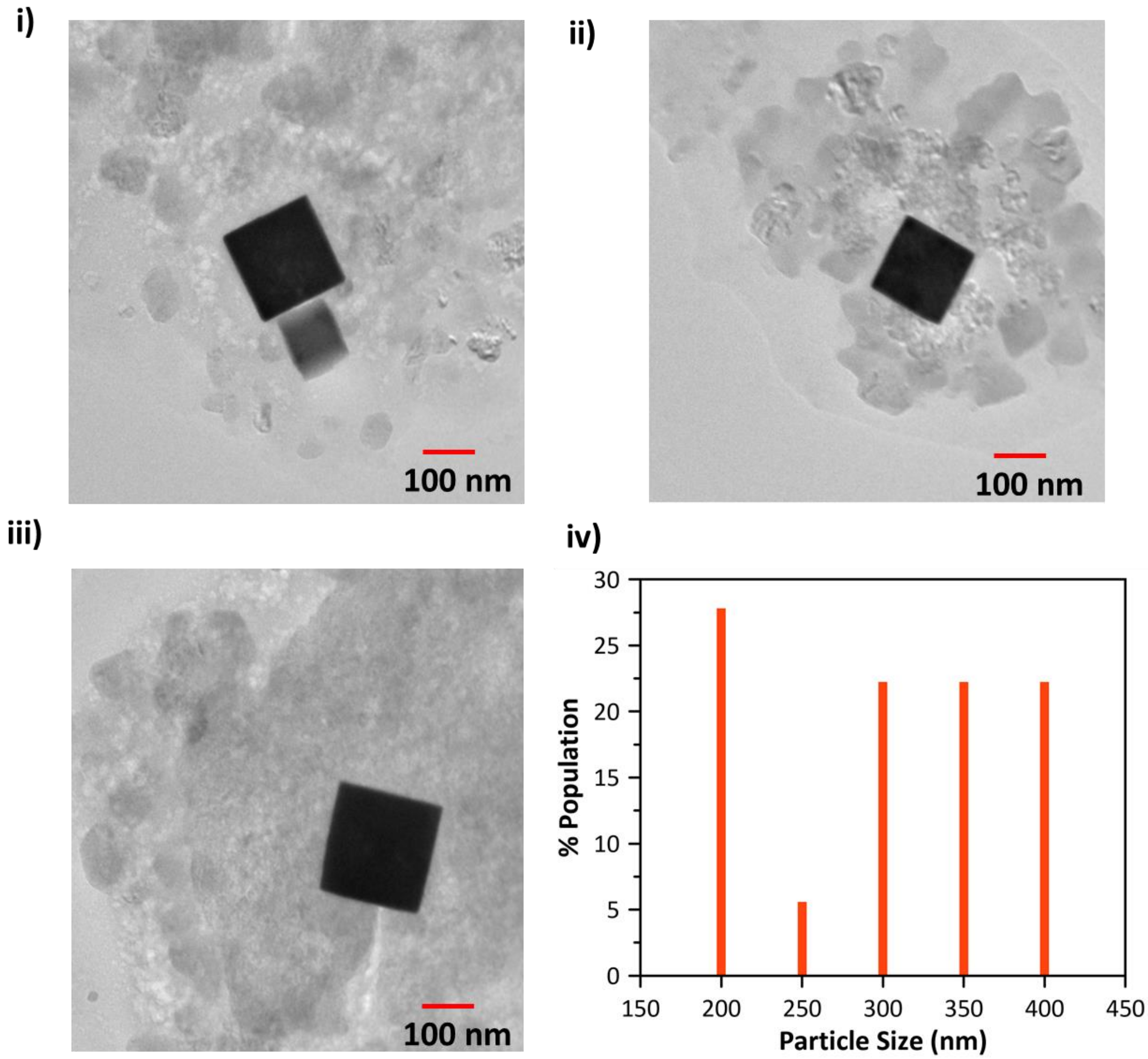

Figure S2f. i-iii) Representative TEM images of particles, and **iv)** size distribution of particles, for the sample collected at 5 minutes during the synthesis of Cu_2O cubes. Average size of particles = 267 ± 83 nm.

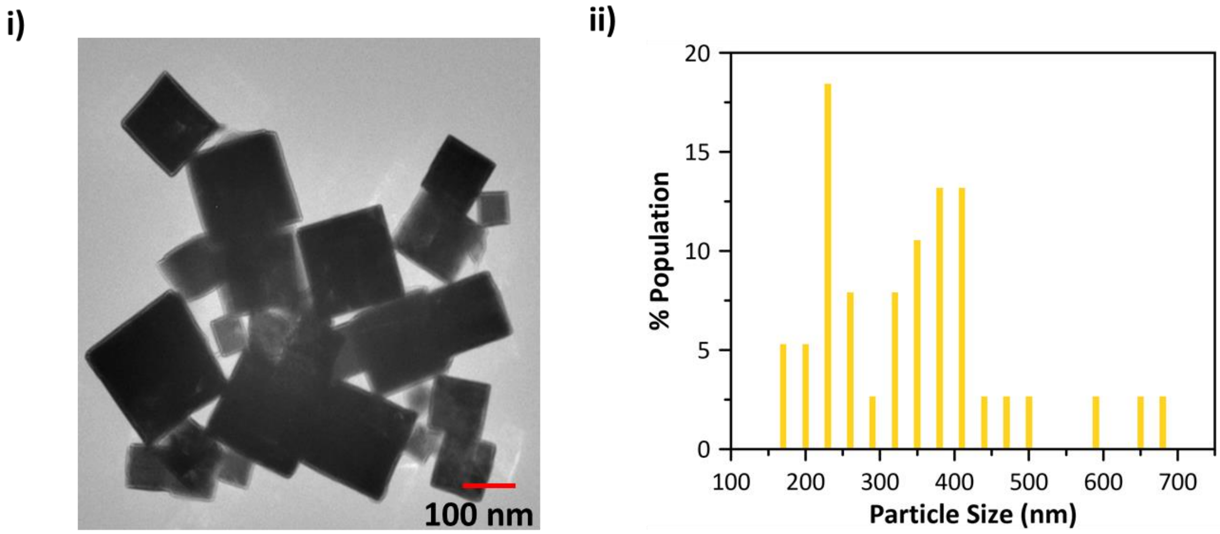

Figure S2g. i) Representative TEM images of particles, and **ii)** size distribution of particles, for the sample collected at 75 minutes during the synthesis of Cu_2O cubes. Average size of particles = 327.3 ±125 nm.

Cu2O spheres:

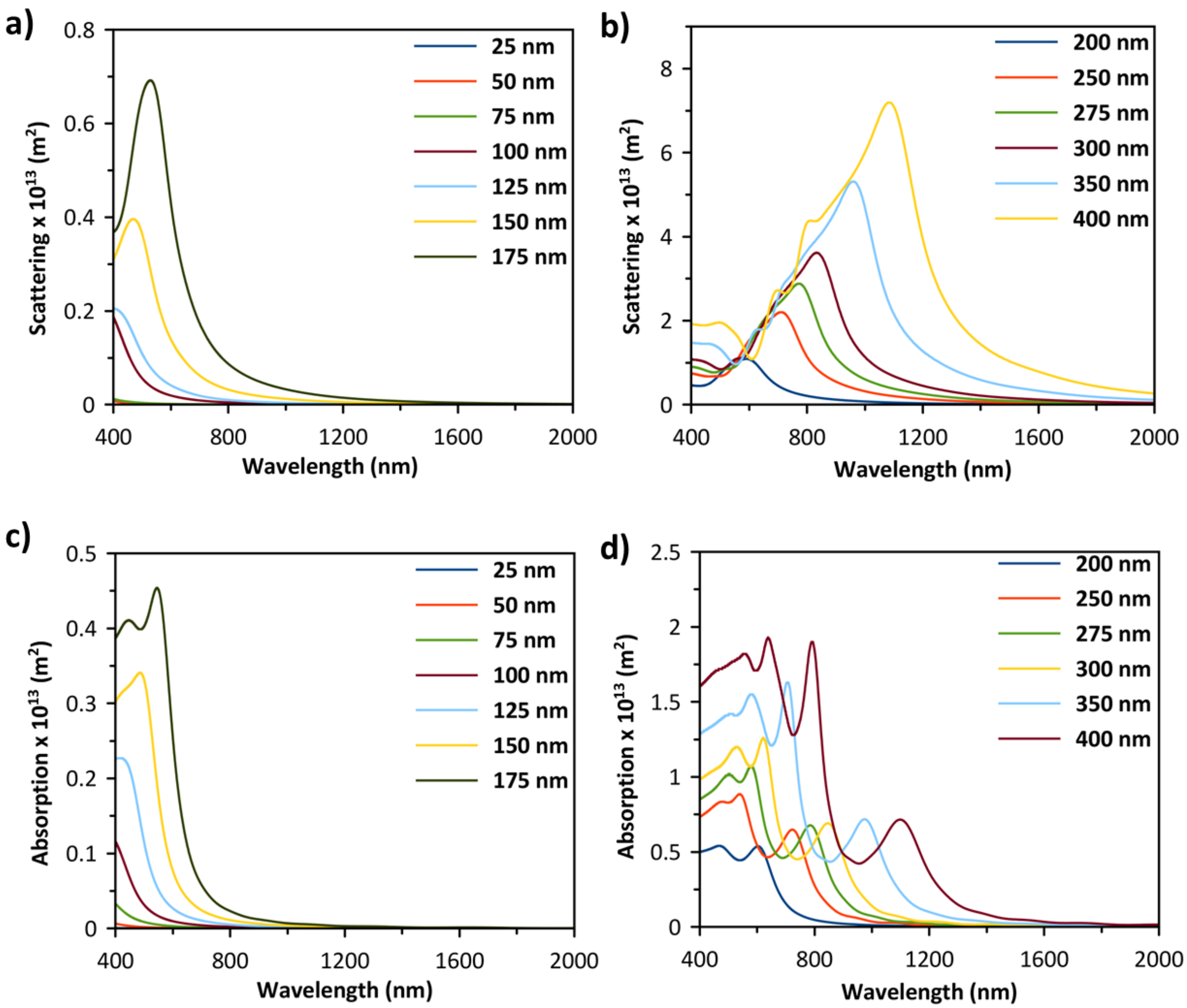

Figure S3. (a and b) Simulated scattering cross section as a function of wavelength for Cu_2O spheres of various diameters ranging from (a) 25 to 175 nm, and (b) 200 to 400 nm. (c and d) Simulated absorption cross section as a function of wavelength for Cu_2O spheres of various diameters ranging from (c) 25 to 175 nm, and (d) 200 to 400 nm.

Cu₂O cubes:

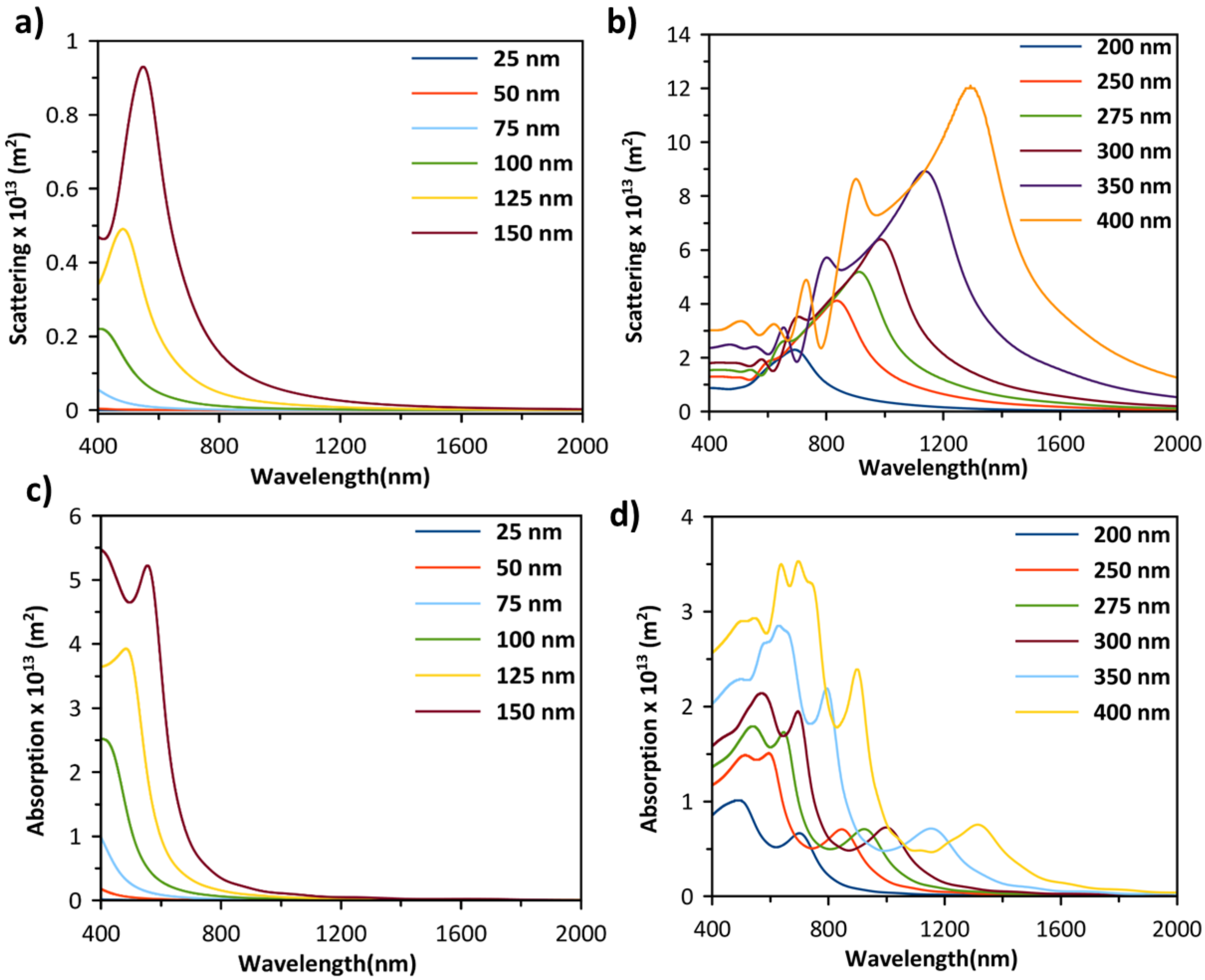

Figure S4. (a and b) Simulated scattering cross section as a function of wavelength for Cu_2O cubes of various edge lengths ranging from (a) 25 to 150 nm, and (b) 200 to 400 nm. (c and d) Simulated absorption cross-section as a function of wavelength for Cu_2O cubes of various edge lengths ranging from (c) 25 to 150 nm, and (d) 200 to 400 nm. The source illumination direction is perpendicular to the principal axis.

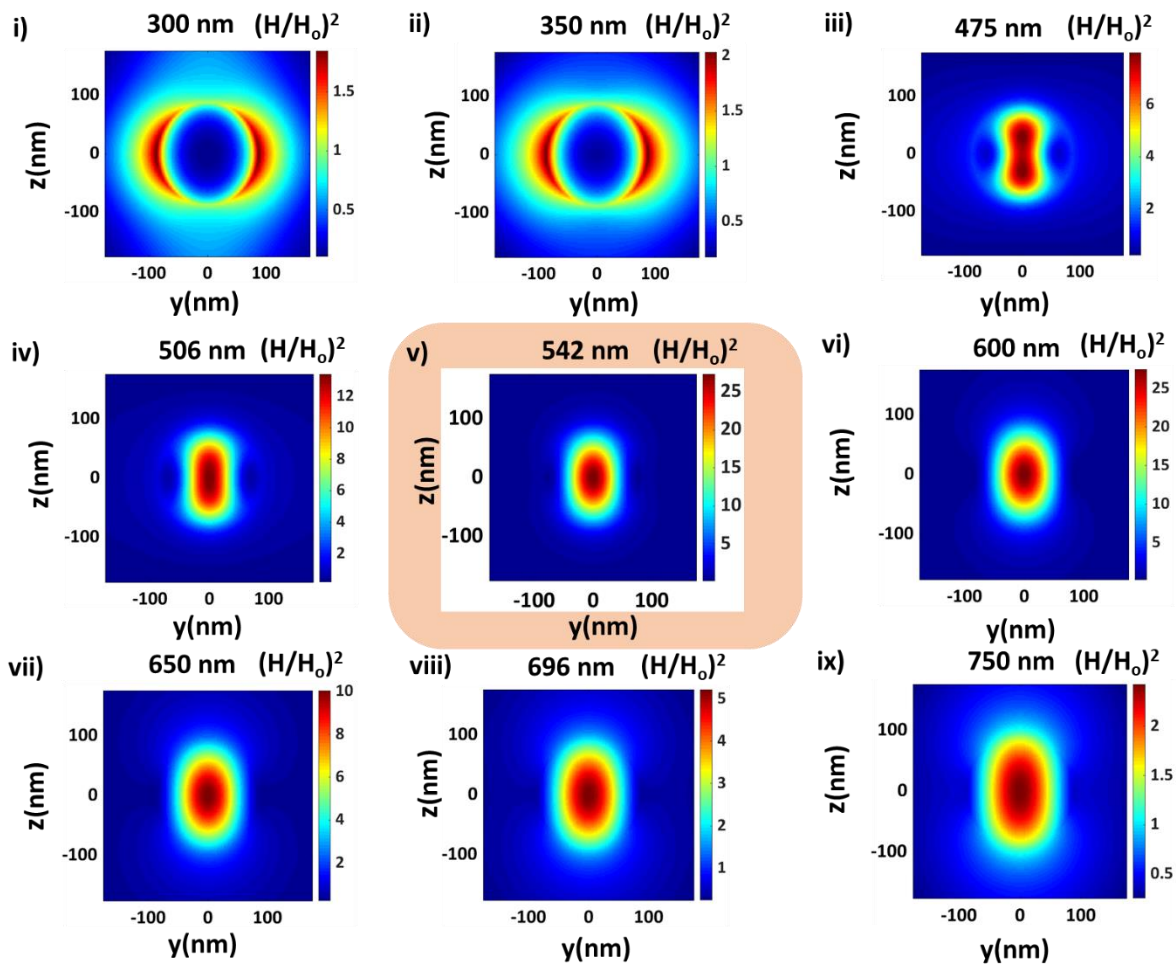

Figure S5a. Simulated spatial distribution of enhancement in magnetic field intensity [H^2/H_0^2] in YZ plane at different wavelengths across the Mie resonance peak wavelength (i.e., **542 nm**) for Cu_2O sphere of 175 nm diameter.

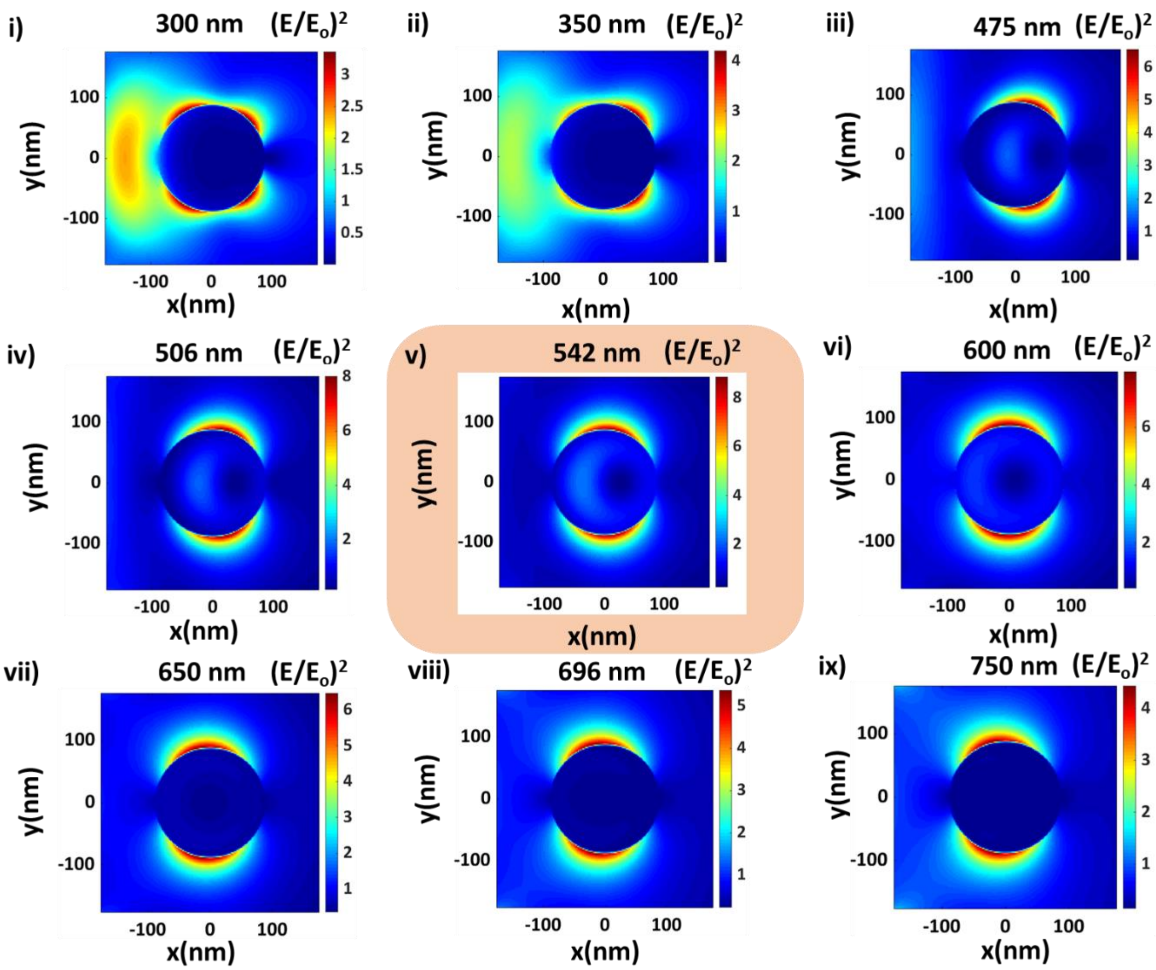

Figure S5b. Simulated spatial distribution of enhancement in electric field intensity [E^2/E_0^2] in XY plane at different wavelengths across the Mie resonance peak wavelength (i.e., **542 nm**) for Cu_2O sphere of 175 nm diameter.

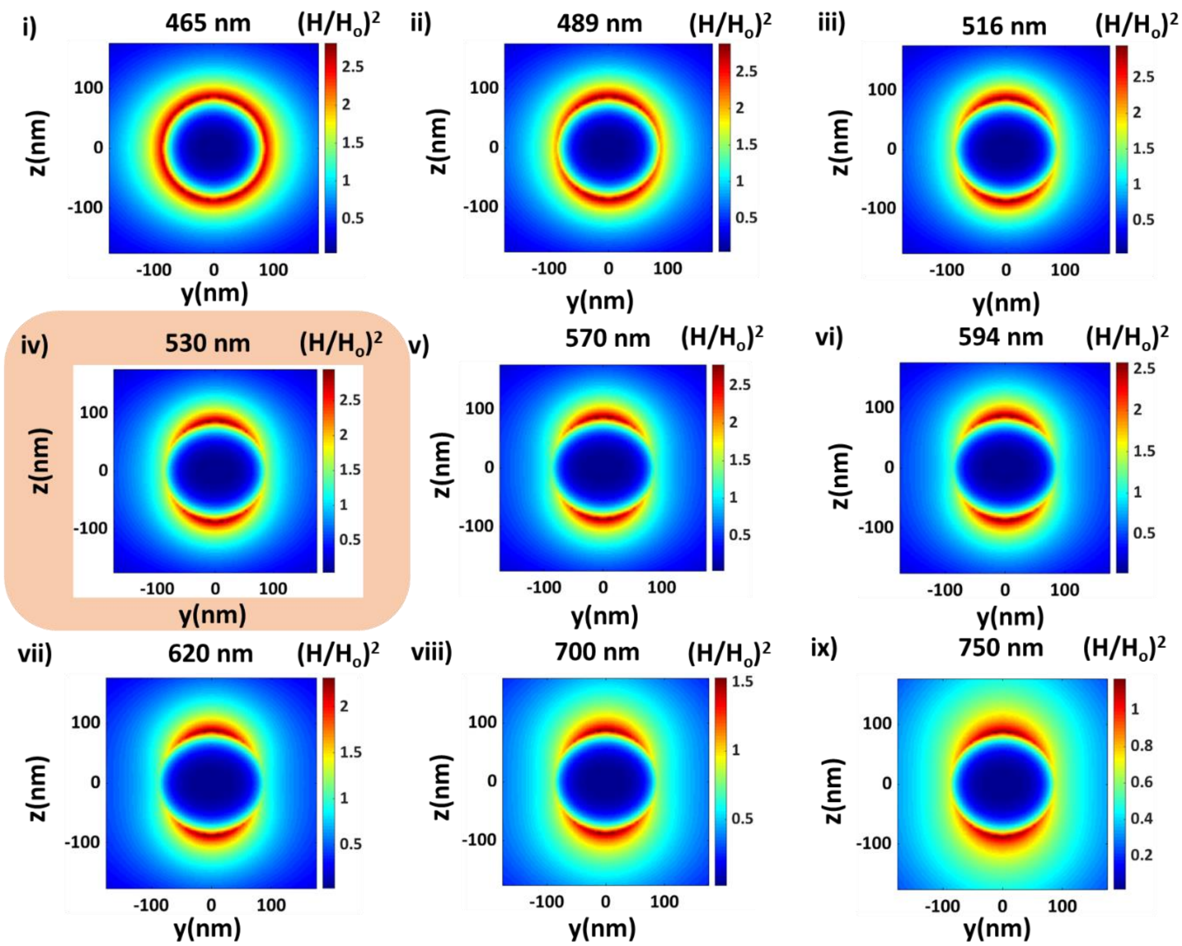

Figure S5c. Simulated spatial distribution of enhancement in magnetic field intensity [H^2/H_0^2] in YZ plane at different wavelengths across the localized surface plasmon resonance (LSPR) peak wavelength (i.e., **530 nm**) for Ag sphere of 175 nm diameter.

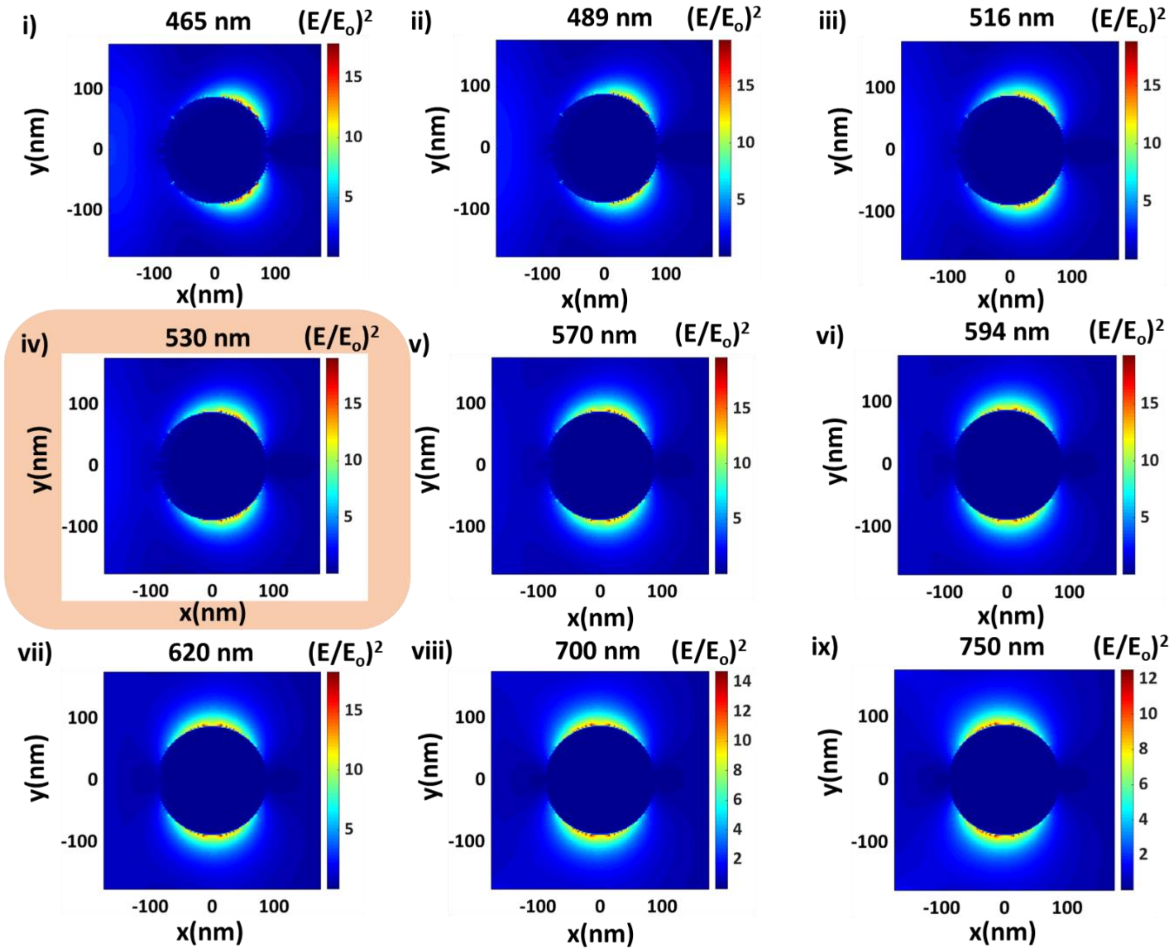

Figure S5d. Simulated spatial distribution of enhancement in electric field intensity [E^2/E_0^2] in XY plane at different wavelengths across the localized surface plasmon resonance (LSPR) peak wavelength (i.e., **530 nm**) for Ag sphere of 175 nm diameter.

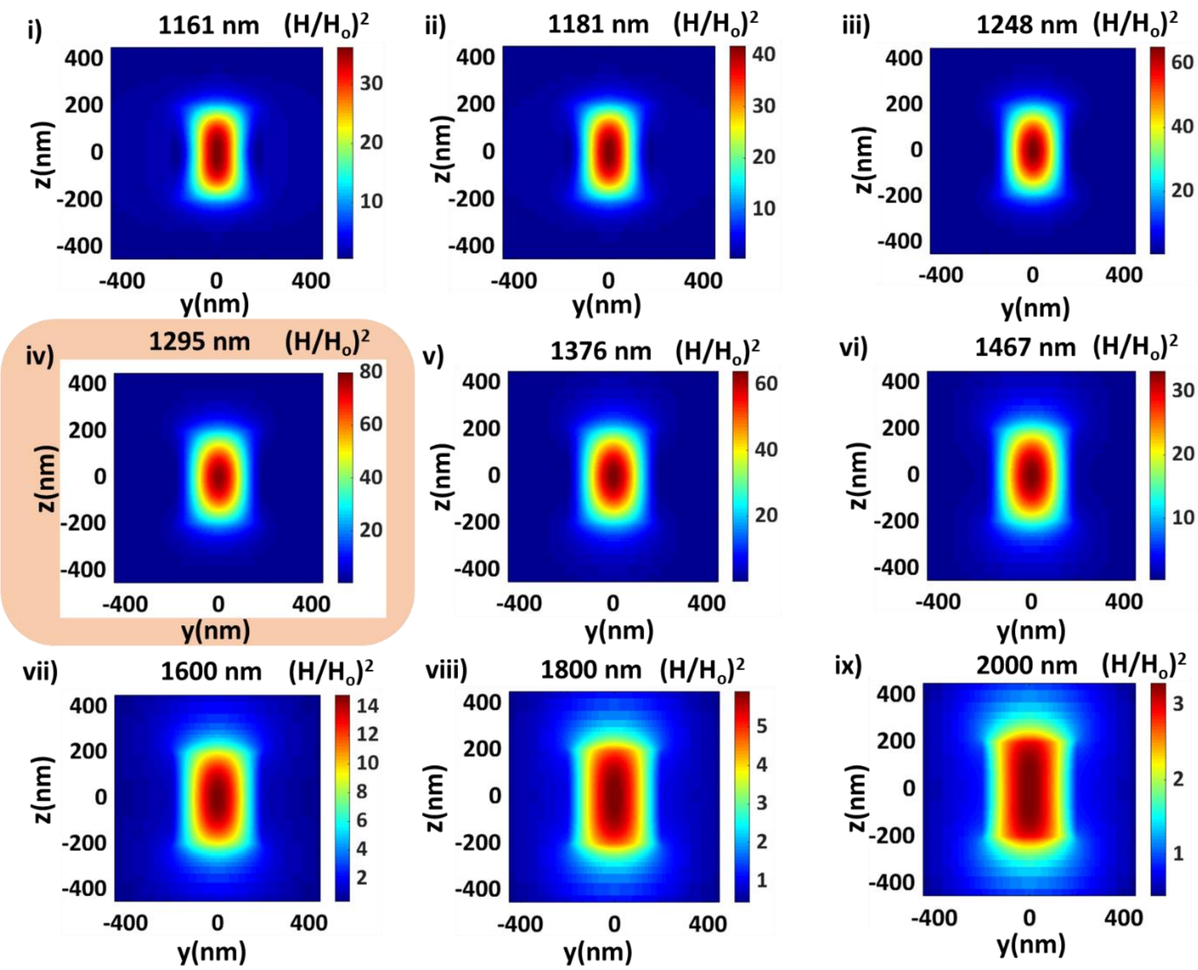

Figure S6a. Simulated spatial distribution of enhancement in magnetic field intensity [H^2/H_0^2] in YZ plane at different wavelengths across the lowest energy Mie resonance peak wavelength (i.e., **1295 nm**) for Cu_2O cube of 400 nm edge length.

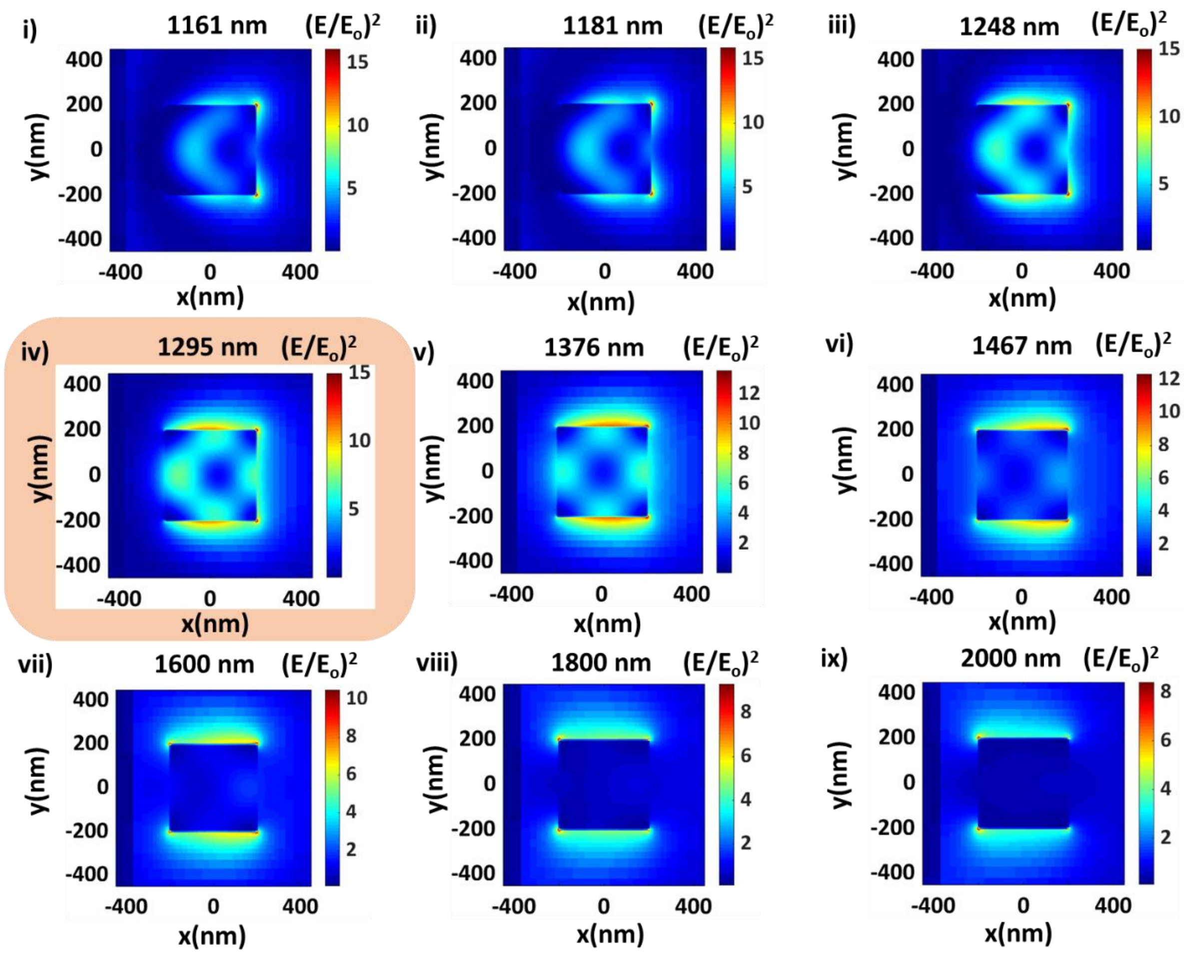

Figure S6b. Simulated spatial distribution of enhancement in electric field intensity [E^2/E_0^2] in XY plane at different wavelengths across the lowest energy Mie resonance peak wavelength (i.e., **1295 nm**) for Cu_2O cube of 400 nm edge length.

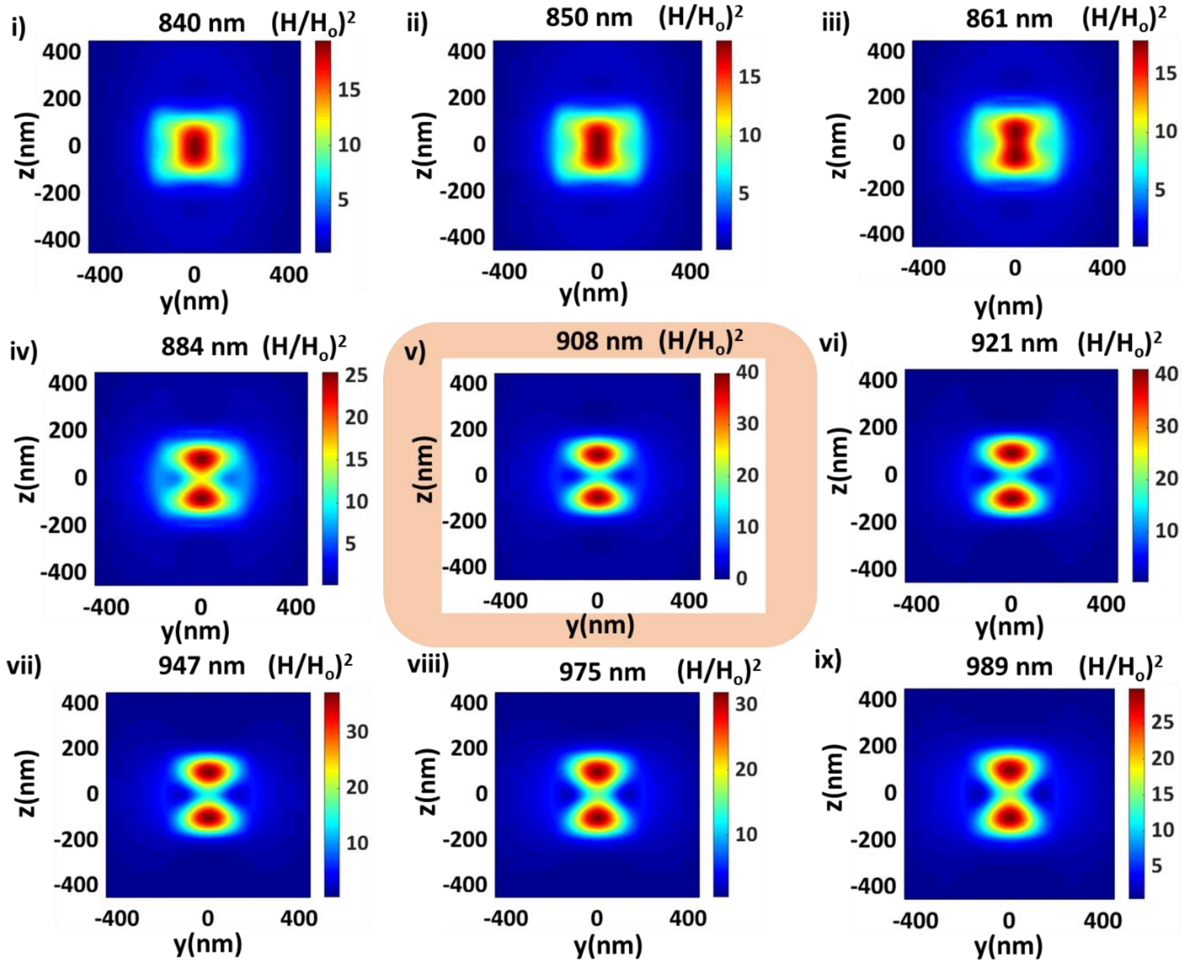

Figure S6c. Simulated spatial distribution of enhancement in magnetic field intensity [H^2/H_0^2] in YZ plane at different wavelengths across the second-lowest energy Mie resonance peak wavelength (i.e., **908 nm**) for Cu_2O cube of 400 nm edge length.

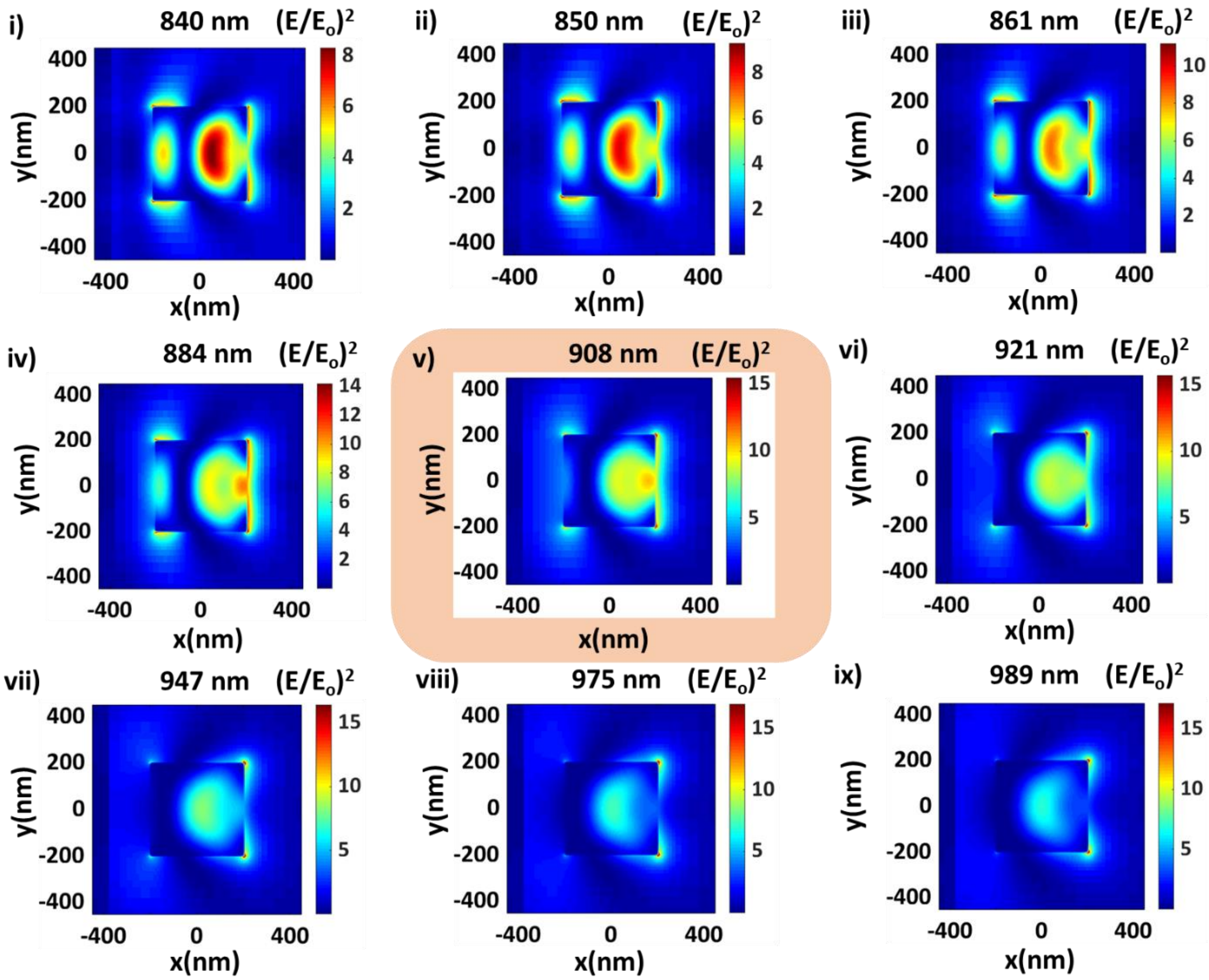

Figure S6d. Simulated spatial distribution of enhancement in electric field intensity [E^2/E_0^2] in XY plane at different wavelengths across the second-lowest energy Mie resonance peak wavelength (i.e., **908 nm**) for Cu_2O cube of 400 nm edge length.

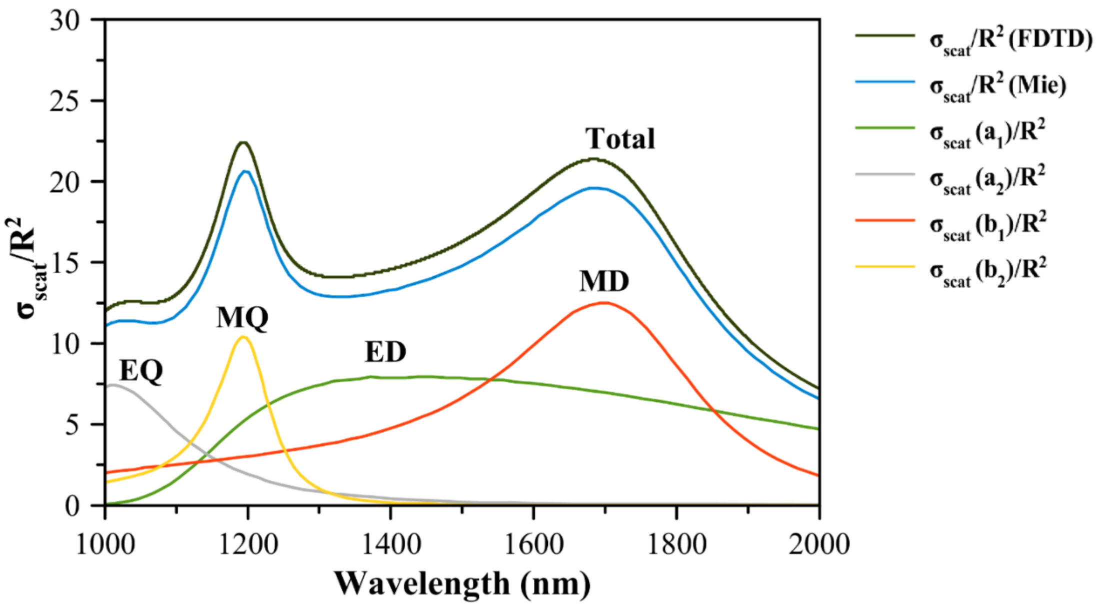

Figure S7. Normalized scattering cross section as a function of wavelength for a lossless dielectric sphere with a constant refractive index of 2.45 and radius (R) of 330 nm. The scattering cross section values (σ_{scat}/R^2, FDTD) simulated from our FDTD simulation matches well with the scattering cross-section values (σ_{scat}/R^2, Mie) reported in the literature using Mie scattering theory.[6] The corresponding Mie coefficient values (a_1, a_2, b_1, and b_2) reported in the literature are also shown.[6] σ_{scat} (a_1), σ_{scat} (a_2), σ_{scat} (b_1), and σ_{scat} (b_2) correspond to scattering contribution from electric dipole (ED), electric quadrupole (EQ), magnetic dipole (MD), and magnetic quadrupole (MQ), respectively. The scattering at the lowest energy Mie resonance peak (i.e., **1714 nm**) is due to the contributions from the magnetic dipole (MD) and electric dipole (ED) resonance modes. At the wavelength of **1714 nm**, the extent of the magnetic dipole contribution is relatively more to the scattering cross-section than the electric dipole. The magnetic dipole (MD) and electric dipole (ED) resonance modes contribute equally to the scattering cross-section at the wavelength of **1548 nm**.

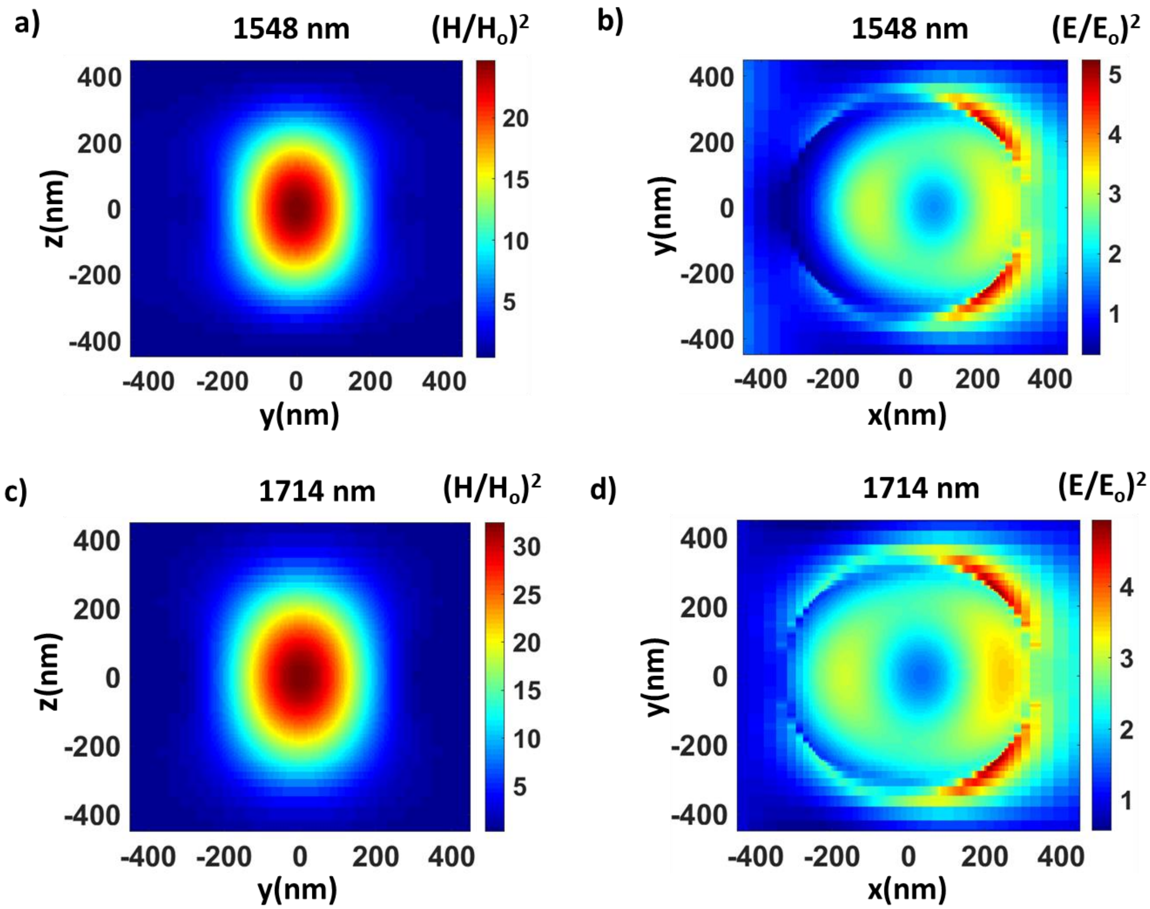

Figure S8. Simulated spatial distribution of enhancement in magnetic and electric field intensity for a dielectric sphere with a constant refractive index of 2.45 and radius (R) of 330 nm. (**a and b**) Simulated spatial distribution of enhancement in (**a**) magnetic field intensity, $[H/H_0]^2$ in YZ plane, and (**b**) electric field intensity, $[E/E_0]^2$ in XY plane at the wavelength of **1548 nm** where the magnetic dipole (MD) and the electric dipole (ED) resonance modes contribute equally to the scattering cross-section. (**c and d**) Simulated spatial distribution of enhancement in (**c**) magnetic field intensity, $[H/H_0]^2$ in YZ plane, and (**d**) electric field intensity, $[E/E_0]^2$ in XY plane at the lowest energy Mie resonance peak wavelength (i.e., **1714 nm**) where the extent of the magnetic dipole contribution is relatively more to the scattering cross-section than the electric dipole.

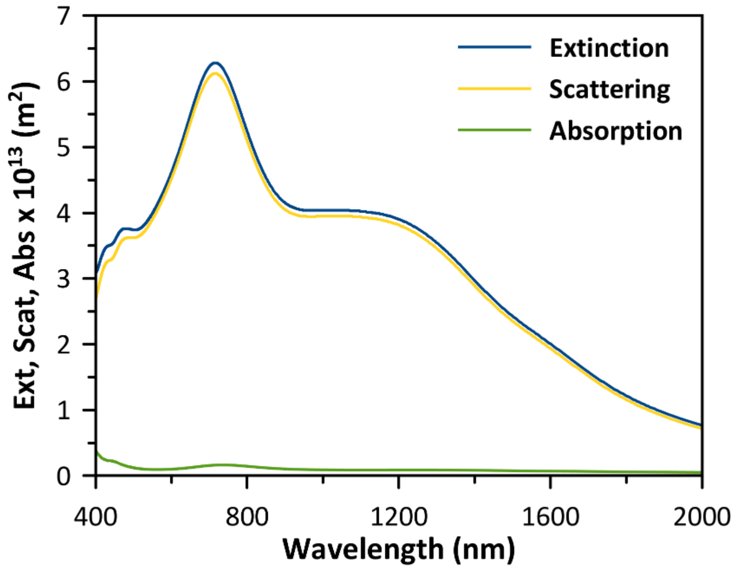

Figure S9. Simulated extinction, scattering, and absorption cross section as a function of wavelength for Ag cube with edge length of 300 nm. The source illumination direction is perpendicular to the principal axis.

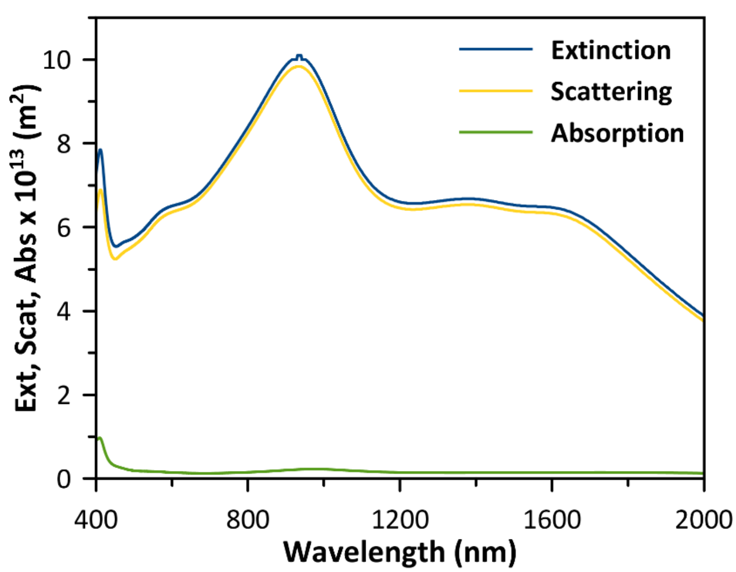

Figure S10. Simulated extinction, scattering, and absorption cross section as a function of wavelength for Ag cube with edge length of 400 nm. The source illumination direction is perpendicular to the principal axis.

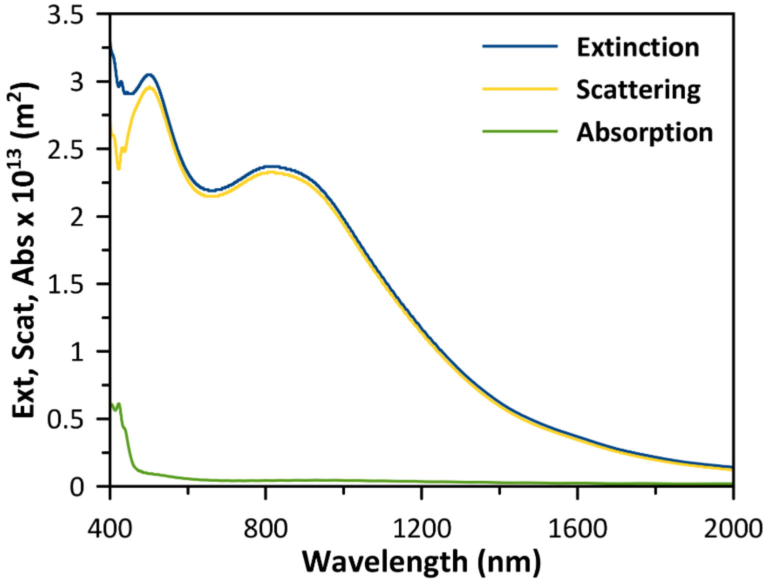

Figure S11. Simulated extinction, scattering, and absorption cross section as a function of wavelength for Ag sphere with diameter of 300 nm.

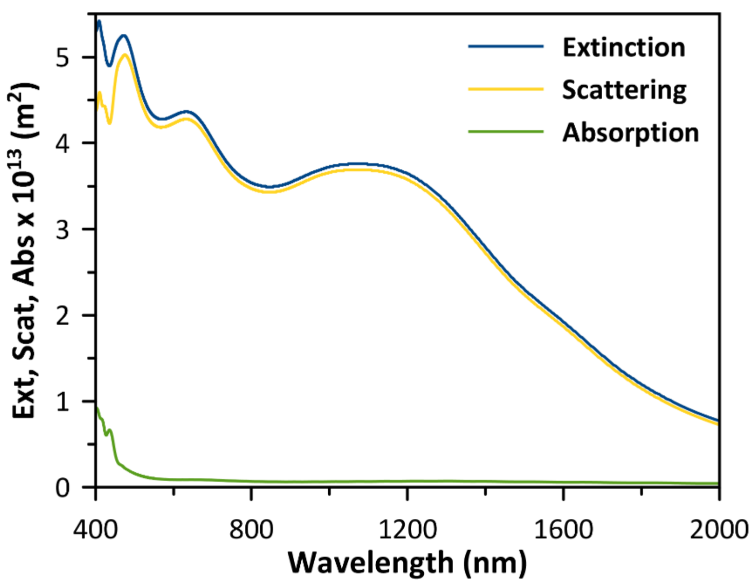

Figure S12. Simulated extinction, scattering, and absorption cross section as a function of wavelength for Ag sphere with diameter of 400 nm.

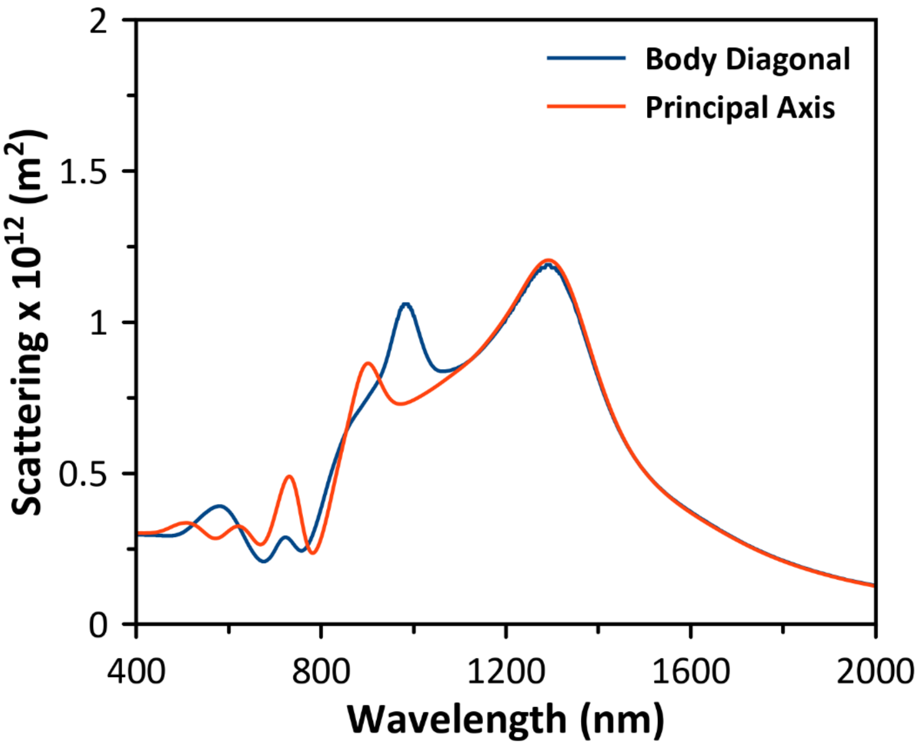

Figure S13. FDTD-simulated scattering cross section values as a function of wavelength for the source light incidence along the body diagonal of Cu_2O cubic particle and the incidence perpendicular to the principal axis of cubic particle. The edge length of particle used in the simulations is 400 nm.

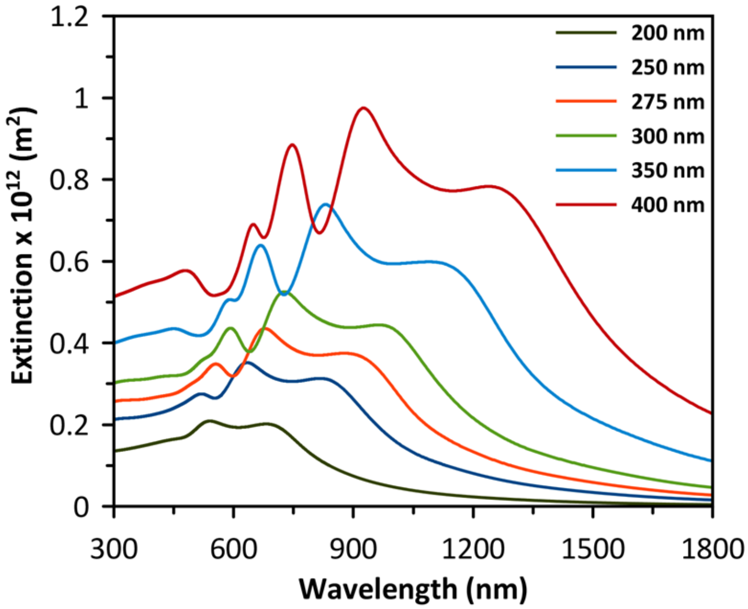

Figure S14. Simulated extinction cross section as a function of wavelength for Cu_2O cubes of various edge lengths ranging from 200 to 400 nm. The source illumination direction is perpendicular to the principal axis. The refractive index of the medium surrounding the cube = 1.33.

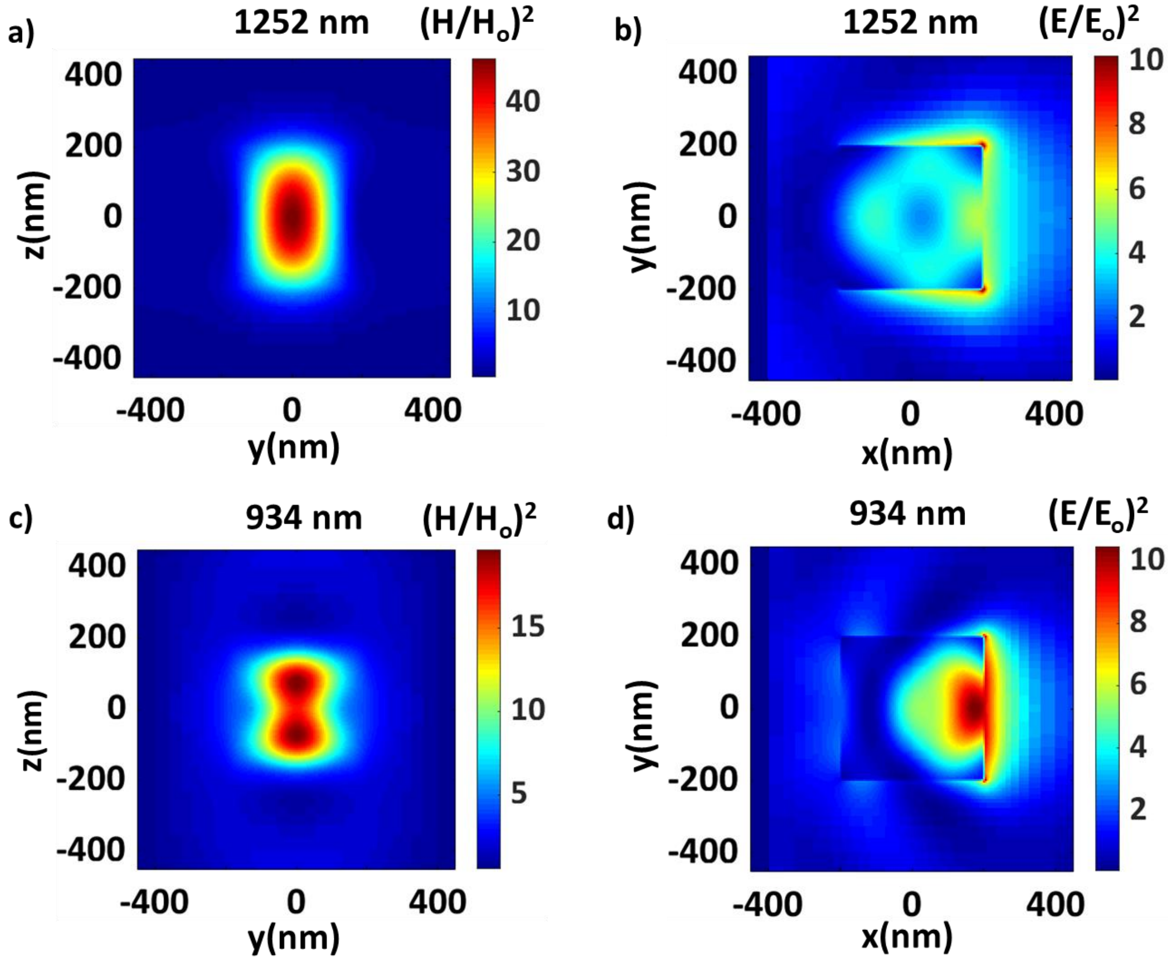

Figure S15. Simulated spatial distribution of enhancement in (a and c) magnetic field intensity [H^2/H_0^2] in YZ plane, and (b and d) electric field intensity [E^2/E_0^2] in XY plane for Cu_2O cubes of 400 nm edge length at the lowest energy Mie resonance wavelength of (a and b) 1252 nm and the second-lowest energy Mie resonance wavelength of (c and d) 934 nm. The refractive index of the medium surrounding the cube = 1.33.

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